

Erdős Problem #970

EP-970

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $h(k)$ be Jacobsthal's function, defined to be the minimal m such that, if n has at most k prime factors, then in any set of m consecutive integers there exists an integer coprime to n . Determine the order of magnitude of $h(k)$. In particular, is it true that $h(k) \ll k^2$?

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsignedMath>